

MyHealthReport

Version 1 Framework — Expanded Edition

Motivation · Method · Result · Conclusion, with schematics for every module

22 August 2026

Agenda

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- 2 System Architecture
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1. Purpose of Version 1

Motivation

Users receive raw lab numbers with no structured way to understand, prioritise, or act on them.

Result

A platform scope statement: interpret, prioritise, explain and track – not diagnose.

Method

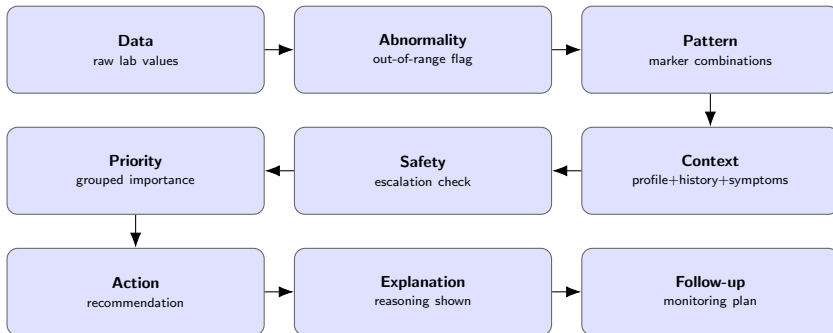
Define nine sequential user-facing objectives, from profile through follow-up, without attempting independent diagnosis.

Conclusion

The Central Logic chain (Data → ... → Follow-up) is the single sentence Version 1 is built to deliver.

1. Purpose of Version 1

The Central Logic, as a flowchart with keywords



2. Core Design Principles

Motivation

Without guiding principles, feature decisions drift toward complexity and clinical overreach.

Result

Five testable rules every later module must satisfy.

Method

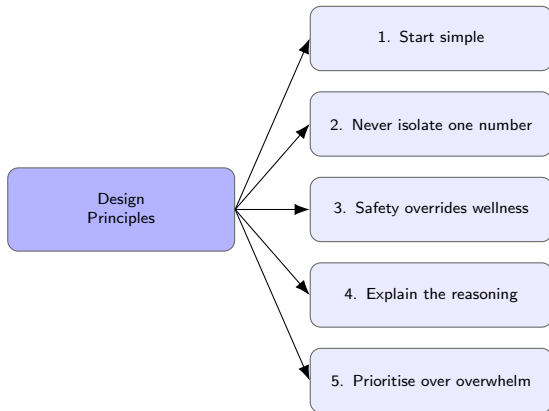
Define five constraints spanning intake, interpretation, safety, transparency and prioritisation.

Conclusion

These principles are the filter every subsequent design decision passes through.

2. Core Design Principles

Five constraints radiating from one design philosophy



3. Version 1 System Architecture

Motivation

A 20-module system needs one map showing how every piece connects before any module is built.

Result

A macro structure of 7 functional clusters, and a micro structure of 22 concrete pipeline stages tagged by module.

Method

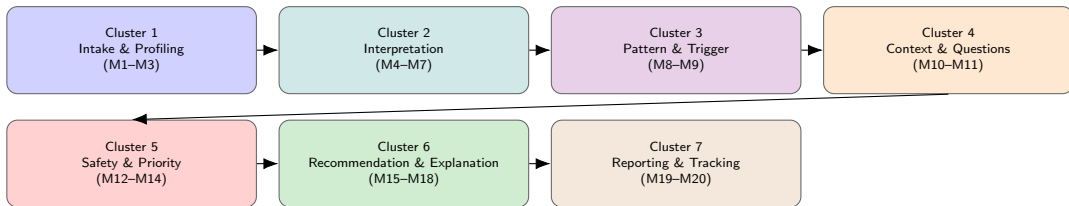
Lay out the full pipeline from registration to trend analysis, then group it by module cluster (macro) and trace it stage-by-stage (micro).

Conclusion

Macro shows *why* modules are grouped; micro shows *exactly* how data moves between them.

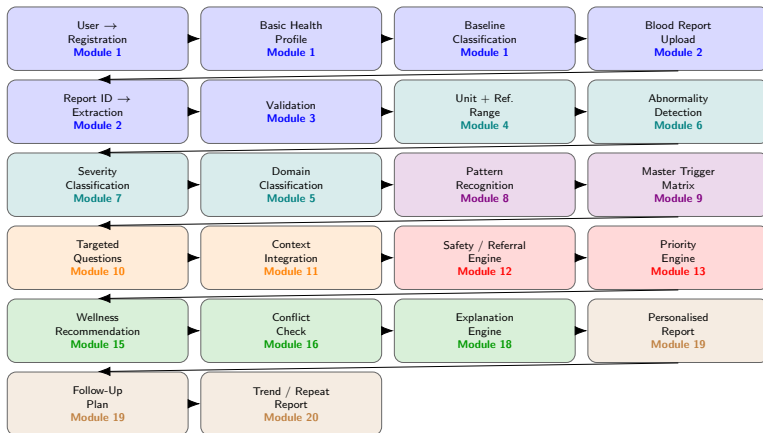
3. Version 1 System Architecture

Macro structure – 7 clusters, module ranges, cluster-to-cluster flow



3. Version 1 System Architecture

Micro structure – 22 pipeline stages, each tagged by its module/cluster



Module 1: Basic Health Profile

Cluster 1 – **Intake & Profiling** (M1–M3)

Motivation

Registration must collect only what is useful across almost every future analysis, without a burdensome intake.

Result

A structured Patient Profile (age, sex, BMI, waist, BP, smoking, alcohol, activity, conditions, medication, pregnancy).

Method

Collect demographic, anthropometric and lifestyle fields once; auto-derive BMI from height and weight and classify weight status.

Conclusion

A lightweight, reusable baseline that every later module references instead of re-asking the same questions.

Schematic 1 – General case (formula & classification) vs. a worked example

GENERAL CASE

$$BMI = \frac{Weight (kg)}{Height (m)^2}$$

Underweight <18.5 Normal 18.5–24.9

Overweight 25–29.9 Obese ≥ 30

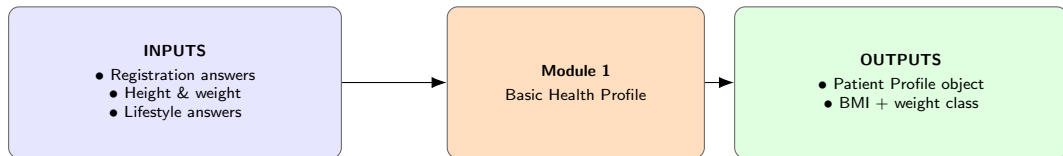
EXAMPLE

Height = 1.70 m Weight = 75 kg

$$BMI = \frac{75}{1.70^2} = 25.95$$

Classification: Overweight

Schematic 2 – Input / Output relations



Motivation

Free-text lab reports cannot be reasoned over; the platform needs structured, comparable data.

Method

Identify report metadata, then extract each result into a fixed record schema (test, result, unit, reference range, flag, date, source).

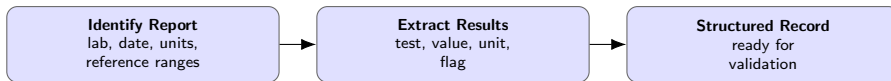
Result

A structured Laboratory Result record for every test on the report.

Conclusion

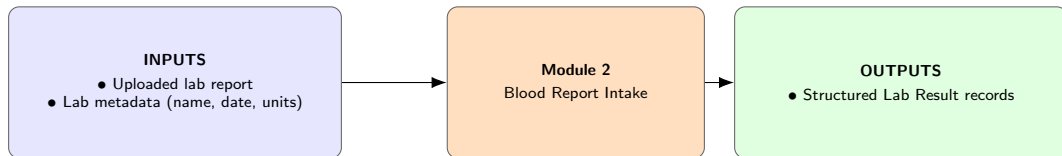
Standardised extraction is the single entry point every later engine ultimately depends on.

Schematic 1 – Two-step intake pipeline



Module 2: Blood Report Intake

Schematic 2 – Input / Output relations



Cluster 1 – **Intake & Profiling** (M1–M3)

Motivation

An extraction error silently becoming health advice is the platform's single biggest safety risk.

Result

Either a validated record proceeds automatically, or the user confirms an uncertain reading first.

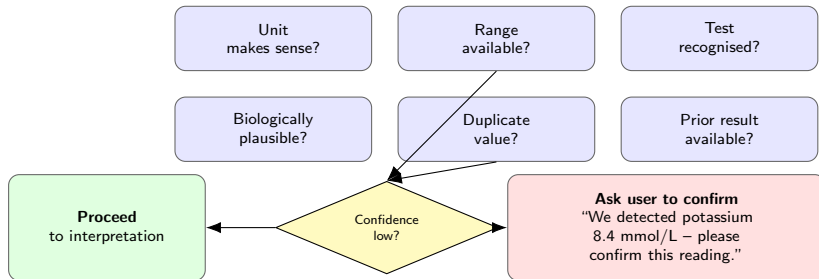
Method

Run plausibility checks on every field before interpretation begins; route low-confidence results to explicit user confirmation.

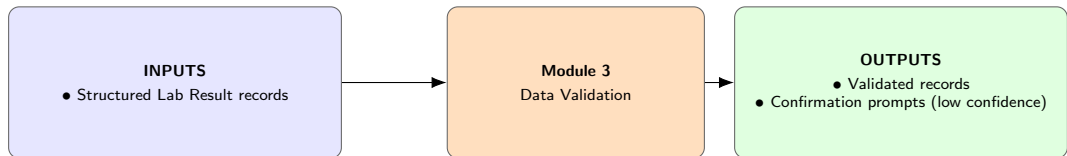
Conclusion

Validation is a gate, not a formality – nothing reaches interpretation unconfirmed.

Schematic 1 – Six checks feed one confidence decision



Schematic 2 – Input / Output relations



Motivation

A single universal reference range cannot serve every lab, guideline and platform rule at once.

Result

A comparable, lab-independent severity signal for every marker (e.g. $2 \times \text{ULN}$).

Method

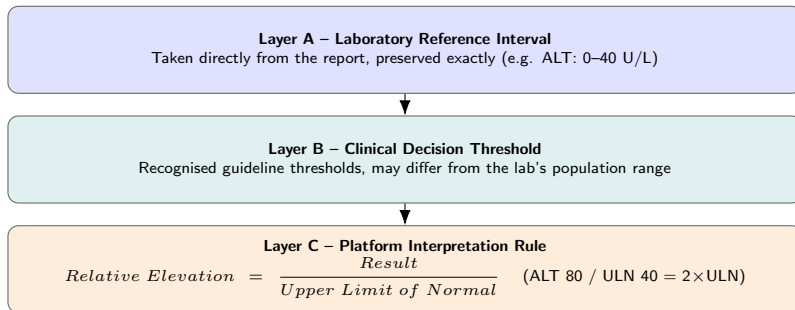
Layer each marker: the lab's own interval preserved exactly, a recognised clinical threshold, and a platform relative-elevation rule.

Conclusion

Layering – not overwriting – keeps original lab data trustworthy while enabling cross-lab comparison.

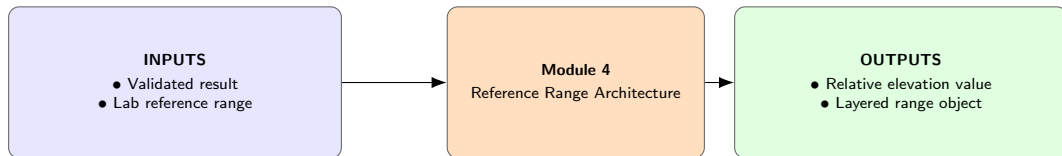
Module 4: Reference Range Architecture

Schematic 1 – Three layers, stacked, not merged



Module 4: Reference Range Architecture

Schematic 2 – Input / Output relations



Cluster 2 – Interpretation (M4–M7)

Motivation

Individual test names mean little without grouping into physiological systems.

Result

Every result is tagged with its domain(s), enabling domain-level review and pattern search.

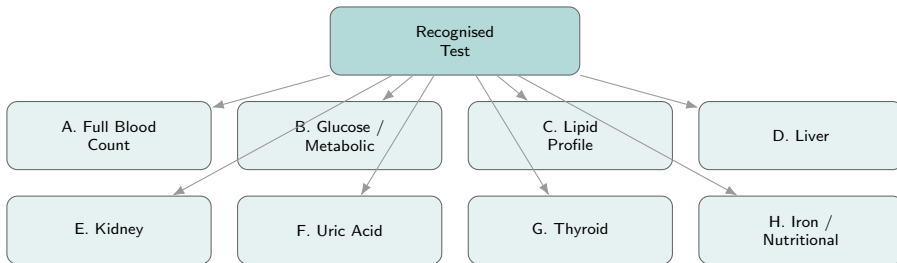
Method

Map each recognised marker into one or more of eight health domains.

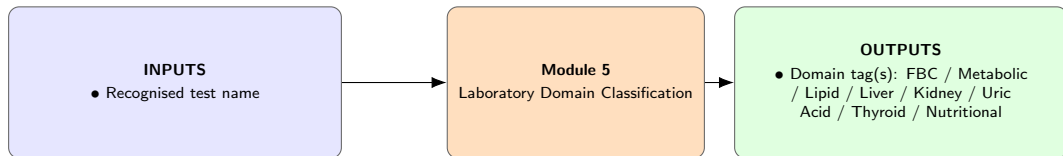
Conclusion

Domain tagging is the scaffold that pattern recognition (Module 8) is built on.

Schematic 1 – One test can route to one or more of eight domains



Schematic 2 – Input / Output relations



Cluster 2 – Interpretation (M4–M7)

Motivation

“Normal / Low / High” alone hides how far outside range a value actually is.

Result

Every result carries both a direction and a graded deviation level.

Method

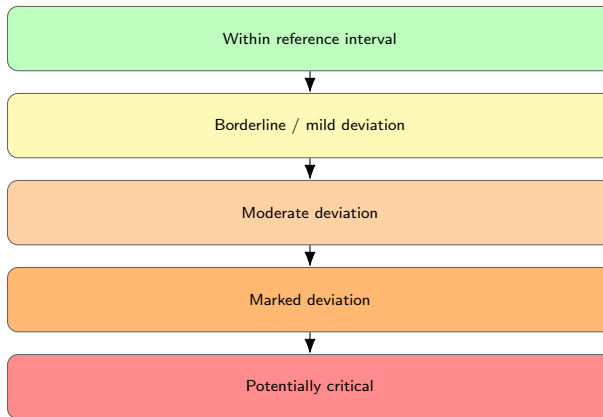
Assign a basic status (Normal/Low/High), then grade magnitude through a five-step deviation scale.

Conclusion

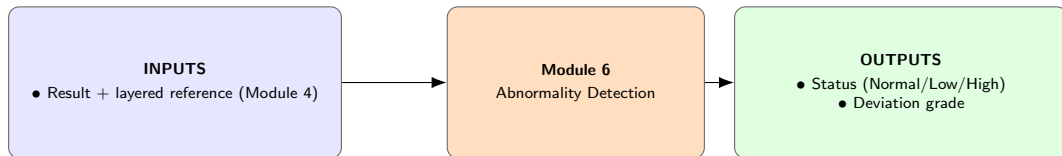
Not all “outside range” values are equally important – magnitude must be visible before severity is assigned.

Module 6: Abnormality Detection

Schematic 1 – Five-step deviation ladder (green → red)



Schematic 2 – Input / Output relations



Cluster 2 – Interpretation (M4–M7)

Motivation

A flat abnormal/normal split cannot route findings to the right next step.

Result

A consistent severity label attached to every finding.

Method

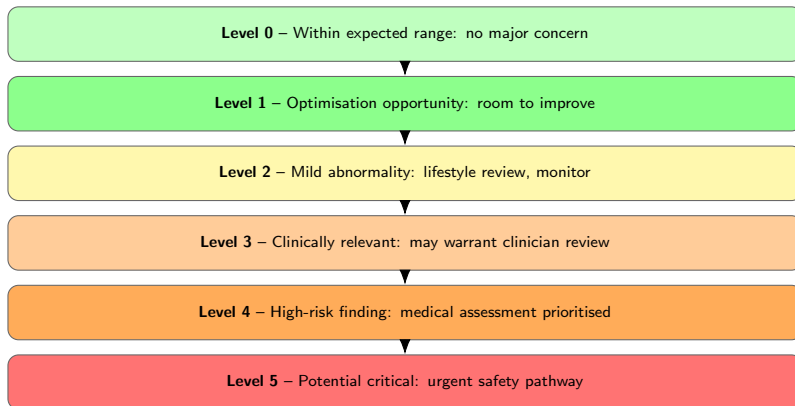
Classify every finding into one of six action levels, from optimisation opportunity to potential critical finding.

Conclusion

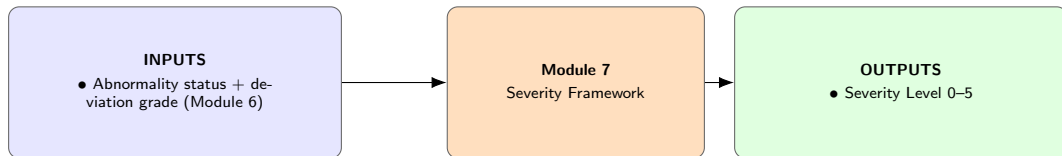
Severity is the shared vocabulary every later engine (questions, safety, priority) reads from.

Module 7: Severity Framework

Schematic 1 – Six action levels, escalating risk



Schematic 2 – Input / Output relations



Cluster 3 – **Pattern & Trigger** (M8–M9)

Motivation

Treating each abnormal value independently misses the clinical picture combinations reveal.

Result

Named patterns – e.g. “metabolic-risk pattern” – attached to the finding set (never a diagnosis).

Method

Search for known marker combinations (metabolic, microcytic, liver) across domains rather than single values.

Conclusion

Patterns turn a list of abnormal numbers into a coherent health narrative.

Schematic 1 – Three worked pattern examples

Metabolic Pattern

BMI↑ waist↑ TG↑ HDL↓
glucose/HbA1c↑ BP↑ ALT↑

Microcytic Pattern

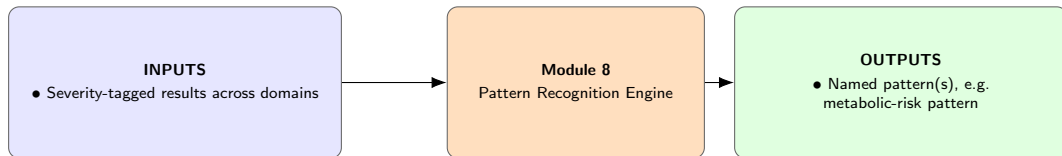
Hb↓ MCV↓ MCH↓
→ ferritin, iron studies,
blood-loss & diet history

Liver Pattern

ALT/AST-predominant
vs. ALP/GGT-predominant
→ different questions

Module 8: Pattern Recognition Engine

Schematic 2 – Input / Output relations



Cluster 3 – Pattern & Trigger (M8–M9)

Motivation

Asking every possible question overwhelms the user; asking none misses context.

Result

A minimal, finding-specific question list per patient.

Method

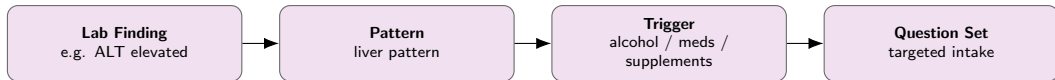
Map each finding/pattern to the specific trigger conditions that activate a targeted question set.

Conclusion

The matrix is the switchboard connecting findings to exactly the right follow-up questions.

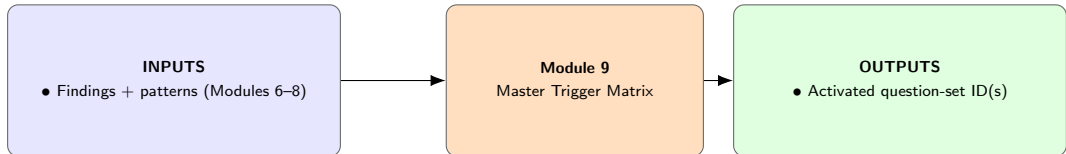
Module 9: Master Trigger Matrix

Schematic 1 – Finding → Pattern → Trigger → Questions



Module 9: Master Trigger Matrix

Schematic 2 – Input / Output relations



Cluster 4 – **Context & Questions** (M10–M11)

Motivation

A full intake questionnaire is unnecessary and burdensome when only specific findings need context.

Result

A short, finding-specific question set the patient actually answers.

Method

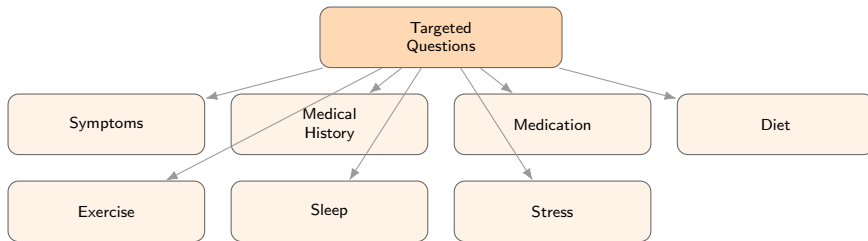
Organise questions into seven categories and activate only the categories a trigger has switched on.

Conclusion

Precision questioning keeps engagement high without sacrificing clinical context.

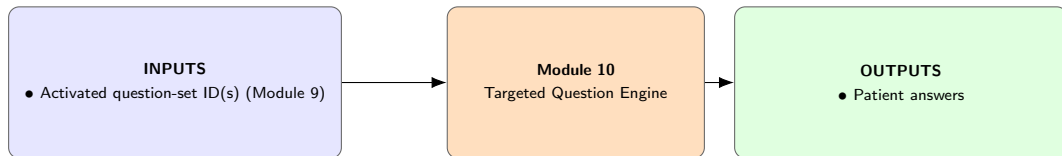
Module 10: Targeted Question Engine

Schematic 1 – Seven question categories, activated selectively



Module 10: Targeted Question Engine

Schematic 2 – Input / Output relations



Module 11: Context Integration Engine

Cluster 4 – **Context & Questions** (M10–M11)

Motivation

A recommendation built from lab data alone ignores the person behind the numbers.

Method

Merge lab results with profile, history, medication, symptoms, lifestyle and previous results into one context object.

Result

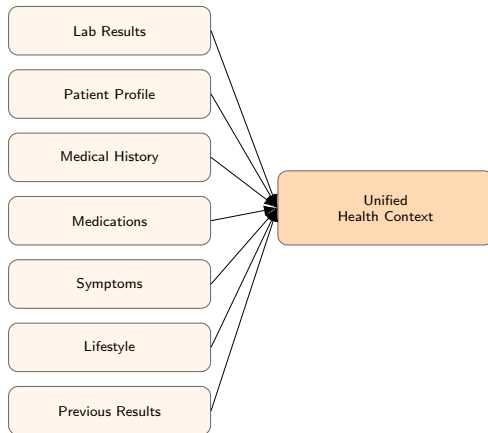
A single, unified Health Context used by every downstream engine.

Conclusion

Integration is what turns isolated data streams into one coherent case.

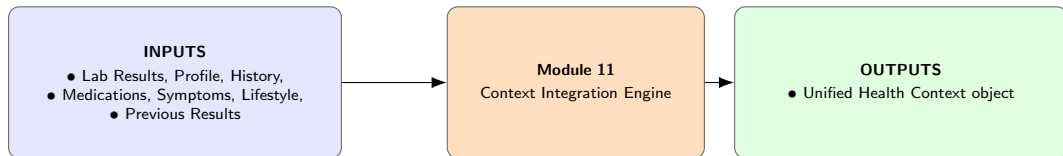
Module 11: Context Integration Engine

Schematic 1 – Seven inputs converge into one context object



Module 11: Context Integration Engine

Schematic 2 – Input / Output relations



Module 12: Safety and Referral Engine

Cluster 5 – **Safety & Priority** (M12–M14)

Motivation

Wellness advice must never mask a finding that needs medical attention.

Result

Every case is routed to Routine Wellness, Medical Review, Prompt Assessment, or Urgent Assessment.

Method

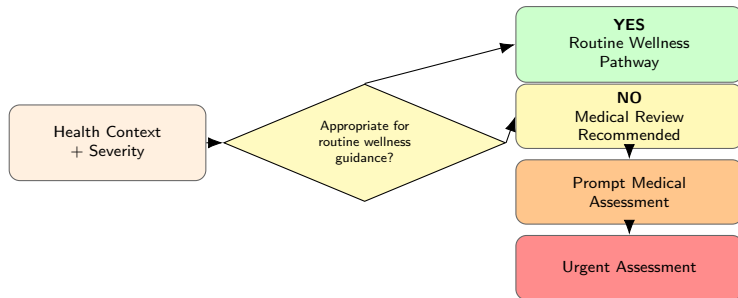
Ask “is this appropriate for routine wellness guidance?” before any recommendation is generated.

Conclusion

Safety runs first and can override everything downstream – by design, not by exception.

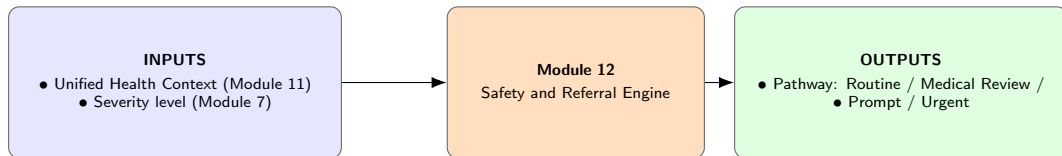
Module 12: Safety and Referral Engine

Schematic 1 – One gate, four possible pathways



Module 12: Safety and Referral Engine

Schematic 2 – Input / Output relations



Cluster 5 – **Safety & Priority** (M12–M14)

Motivation

Six abnormal findings should not produce six competing recommendations.

Result

A short, ranked priority list (typically 3–4 items) instead of a flat finding list.

Method

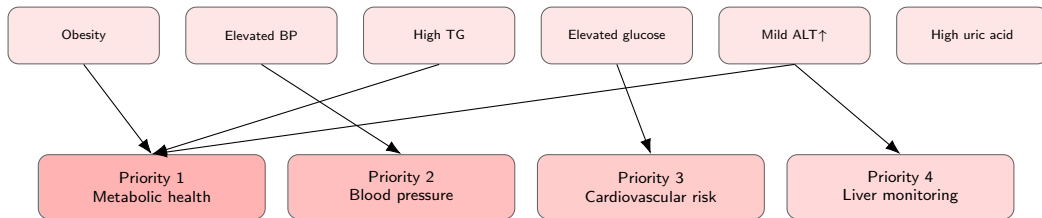
Group related findings and identify the upstream priorities that explain the most downstream issues.

Conclusion

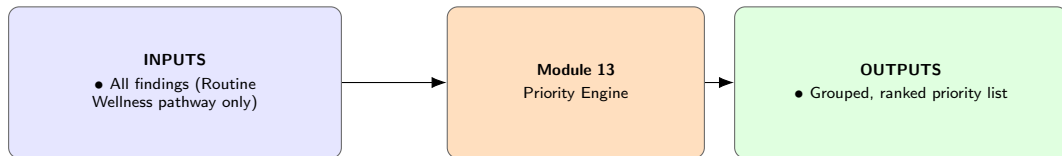
Grouping before recommending is what keeps the final report usable.

Module 13: Priority Engine

Schematic 1 – Many findings converge into a few priorities



Schematic 2 – Input / Output relations



Cluster 5 – **Safety & Priority** (M12–M14)

Motivation

Treating five metabolic abnormalities as five separate problems obscures their shared cause.

Result

A causal chain the report can state explicitly to the user.

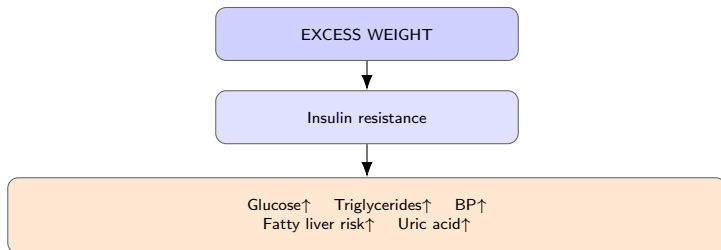
Method

Trace findings back to modifiable upstream factors (e.g. excess weight → insulin resistance → multiple markers).

Conclusion

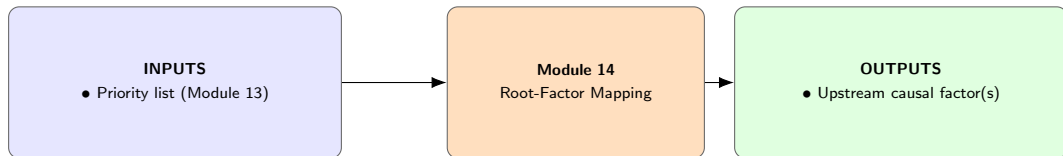
One well-chosen intervention can be shown to improve several findings at once.

Schematic 1 – One upstream factor, five downstream effects



Module 14: Root-Factor Mapping

Schematic 2 – Input / Output relations



19. Recommendation Structure

Motivation

A recommendation without a fixed schema will vary in quality and completeness across cases.

Result

A worked example (triglycerides + weight) showing the schema fully populated.

Method

Require seven fields on every recommendation: trigger, health issue, recommendation, why, expected benefit, precaution, follow-up.

Conclusion

This schema is what Modules 15–18 all read from and write to.

19. Recommendation Structure

Seven-field recommendation schema, with a worked example



Worked example: Trigger = elevated TG + excess weight → Recommendation = reduce refined carbs/sugary drinks, gradual weight loss → Why = excess energy/refined-carb intake raises TG → Benefit = improved TG & metabolic health → Precaution = modify per conditions/medication → Follow-Up = repeat measurement per clinical plan.

Cluster 6 – **Recommendation & Explanation** (M15–M18)

Motivation

Recommendations need a bounded, evidence-proportionate scope for Version 1.

Result

A candidate recommendation set tied to specific triggers.

Method

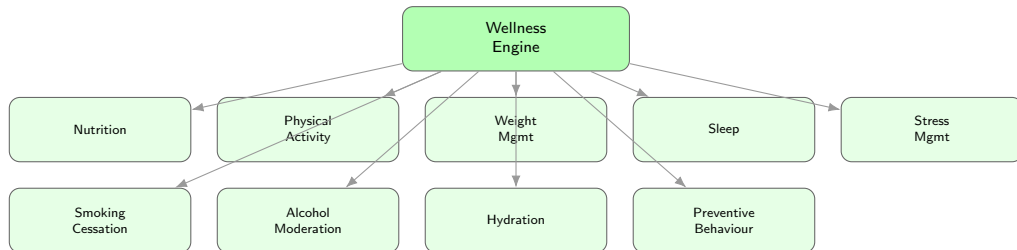
Draw candidate recommendations from nine lifestyle domains, proportional to the evidence available.

Conclusion

Scope discipline keeps Version 1 focused on what wellness guidance can responsibly claim.

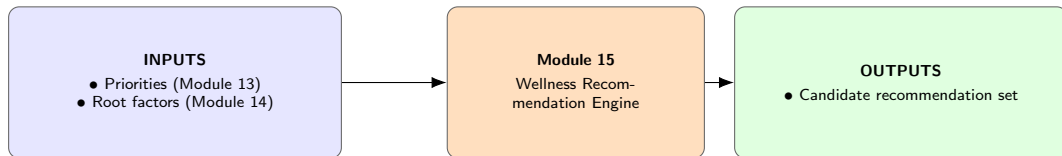
Module 15: Wellness Recommendation Engine

Schematic 1 – Nine bounded recommendation domains



Module 15: Wellness Recommendation Engine

Schematic 2 – Input / Output relations



Cluster 6 – Recommendation & Explanation (M15–M18)

Motivation

A generically sound recommendation can be actively harmful in the wrong patient context.

Result

Recommendations are suppressed or modified whenever a contraindication applies.

Method

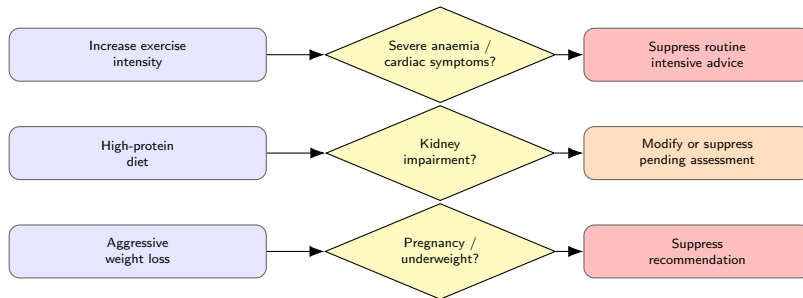
Pass every candidate recommendation through known conflict rules before it is displayed.

Conclusion

No recommendation reaches the user without first surviving a safety check.

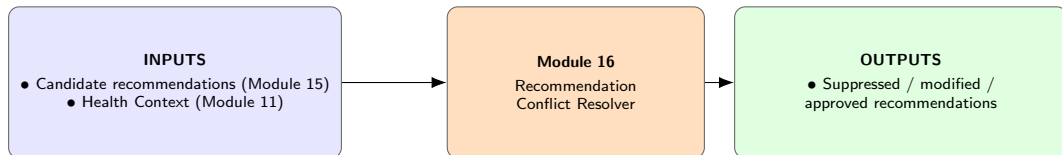
Module 16: Recommendation Conflict Resolver

Schematic 1 – Three worked conflict-check examples



Module 16: Recommendation Conflict Resolver

Schematic 2 – Input / Output relations



Module 17: Recommendation Priority System

Cluster 6 – **Recommendation & Explanation** (M15–M18)

Motivation

A patient handed ten equally-weighted actions acts on none of them.

Result

Roughly three main priorities, with secondary items underneath.

Method

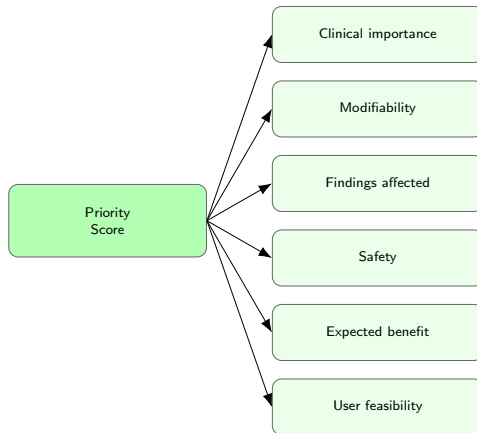
Score each surviving recommendation on six factors: importance, modifiability, findings affected, safety, benefit, feasibility.

Conclusion

Scoring turns a candidate list into a decision the patient can actually act on.

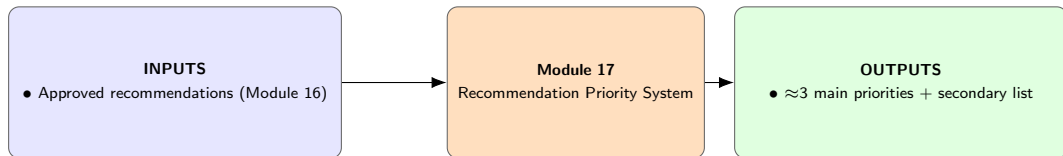
Module 17: Recommendation Priority System

Schematic 1 – Six factors feed one priority score



Module 17: Recommendation Priority System

Schematic 2 – Input / Output relations



Cluster 6 – Recommendation & Explanation (M15–M18)

Motivation

A recommendation without reasoning invites distrust and non-adherence.

Result

A fully reasoned explanation the user can read start-to-finish.

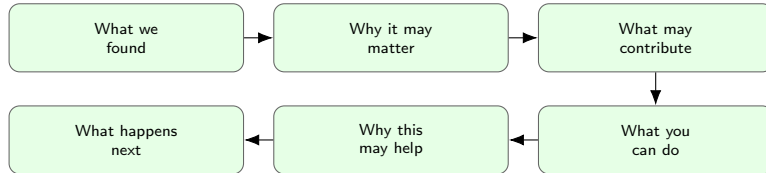
Method

Attach six structured fields to every recommendation, from what was found to what happens next.

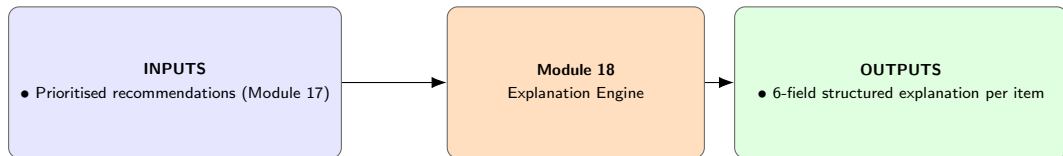
Conclusion

Explaining the “why” is what makes the platform feel like guidance, not a black box.

Schematic 1 – Six-step explanation sequence



Schematic 2 – Input / Output relations



Cluster 7 – Reporting & Tracking (M19–M20)

Motivation

Findings, patterns and recommendations need one coherent document, not scattered outputs.

Result

A single, navigable report the user receives after every analysis.

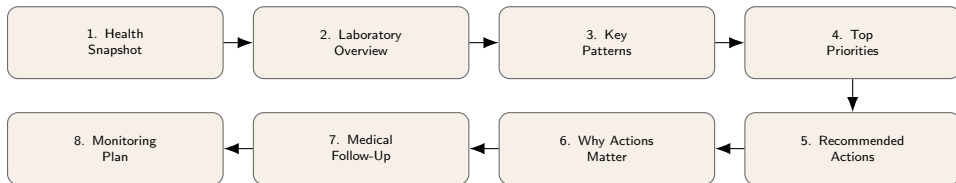
Method

Assemble eight fixed sections, from Health Snapshot through Monitoring Plan, in a consistent order.

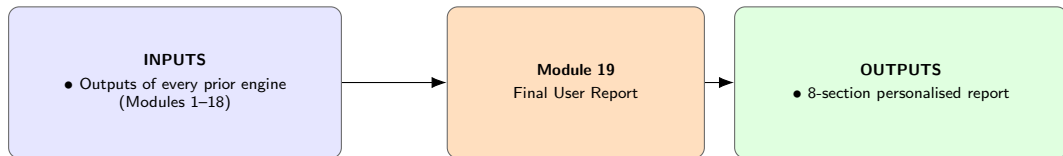
Conclusion

Consistent structure is what makes repeat reports comparable over time.

Schematic 1 – Eight sections, fixed reading order



Schematic 2 – Input / Output relations



Cluster 7 – Reporting & Tracking (M19–M20)

Motivation

A single time-point result cannot show whether a person's health is improving.

Result

Trend-aware statements (e.g. "progressive upward trend across three measurements") instead of isolated snapshots.

Method

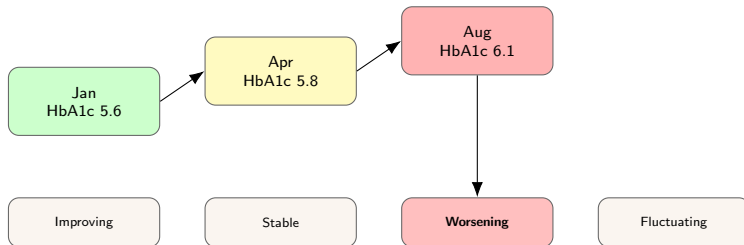
Store every result against its date and classify multi-point series into a trend category.

Conclusion

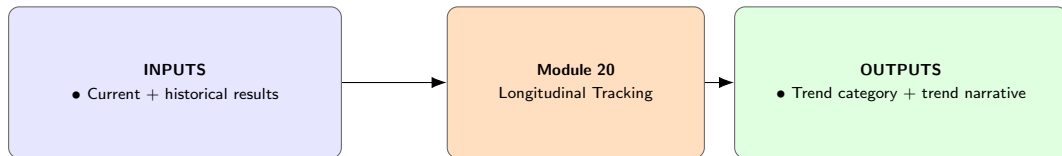
Tracking is what turns MyHealthReport from a one-off reading into an ongoing health record.

Module 20: Longitudinal Tracking

Schematic 1 – A rising HbA1c series classified as Worsening



Schematic 2 – Input / Output relations



25. Version 1 Database Concept

Motivation

Free-text storage cannot support rules, triggers, or trend analysis.

Result

A relational data model underlying every module in the pipeline.

Method

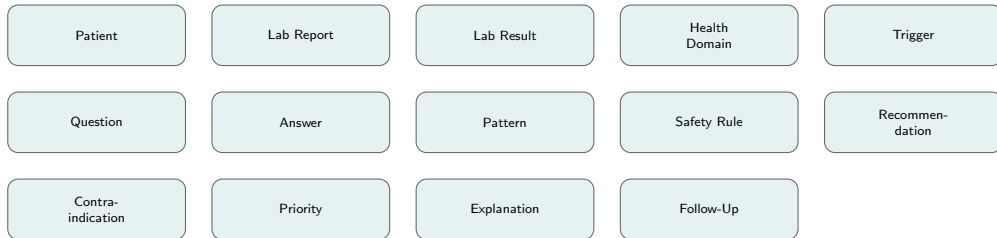
Define 13 core structured objects covering patient data, findings, rules and recommendations.

Conclusion

Every module in this deck reads or writes one or more of these 13 objects.

25. Version 1 Database Concept

Thirteen core structured objects



26. Master Rule Architecture

Motivation

Twenty modules need one shared rule shape, or the logic becomes unmaintainable.

Result

A single reusable rule template applied across all 20 modules.

Method

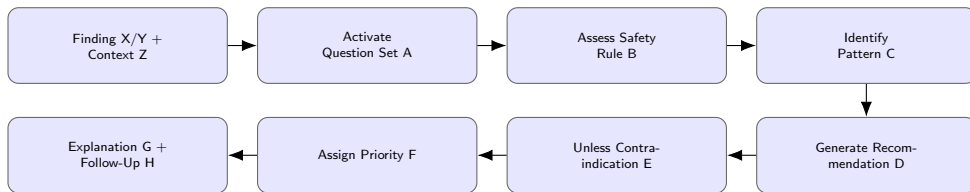
Express every clinical/wellness rule as: IF Finding + Context THEN Question → Safety → Pattern → Recommendation UNLESS Contraindication THEN Priority + Explanation + Follow-Up.

Conclusion

This is the syntax the Four Master Matrices (Topic 31) are written in.

26. Master Rule Architecture

The IF...THEN...UNLESS rule template as a flow



27. What AI Should Do

Motivation

AI is capable of more than the platform should let it decide unsupervised.

Result

A bounded, non-clinical role for AI within the pipeline.

Method

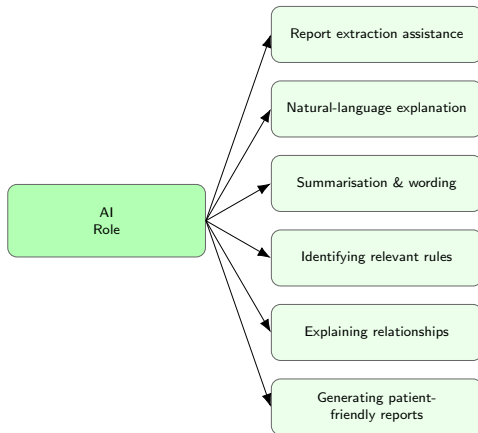
Scope AI to extraction assistance, explanation, summarisation, wording, rule identification and report generation.

Conclusion

AI accelerates communication; it does not originate clinical judgment.

27. What AI Should Do

Six permitted AI functions, radiating from one bounded role



28. What AI Should NOT Independently Control

Motivation

Letting AI freely decide safety-critical logic risks an unvalidated claim reaching a user.

Result

A hard boundary between “rules decide” and “AI explains”.

Method

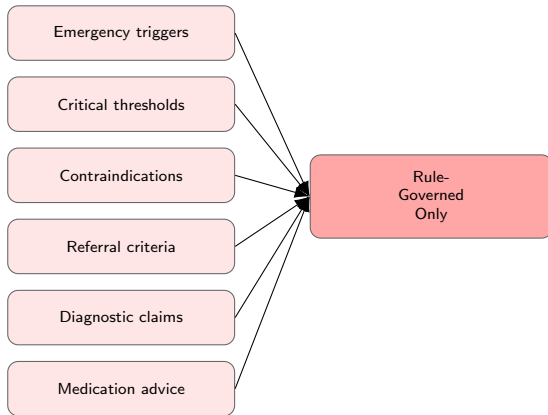
Reserve emergency triggers, thresholds, contraindications, referral criteria, diagnosis and medication advice to rule-governed logic only.

Conclusion

Structured clinical rules determine what is allowed; AI determines how it is explained.

28. What AI Should NOT Independently Control

Six safety-critical elements, reserved to rule-governed logic



29. Version 1 Scope Boundary

Motivation

An unbounded roadmap delays shipping a safe, useful Version 1.

Result

A concrete in/out list reviewable against every new feature request.

Method

Explicitly separate what Version 1 includes from what is delayed.

Conclusion

Saying no to Phase-2+ features is what keeps Version 1 shippable and safe.

29. Version 1 Scope Boundary

Include vs. delay – a reviewable boundary

INCLUDE

Basic profile, BMI, BP context, common blood tests, abnormality detection, severity classification, common pattern recognition, targeted questions, wellness recommendations, red-flag/referral logic, explanations, trend tracking, report generation

DELAY TO LATER VERSIONS

Supplement marketplace, automatic supplement prescribing, pharmacy sales integration, complex genetic interpretation, imaging interpretation, diagnosis engine, medication initiation/adjustment, complex specialist disease management, autonomous AI clinical decisions

30. Clinical Domains to Build First

Motivation

Building all eight lab domains simultaneously multiplies validation risk before any value ships.

Result

A four-phase build order, not a simultaneous eight-domain launch.

Method

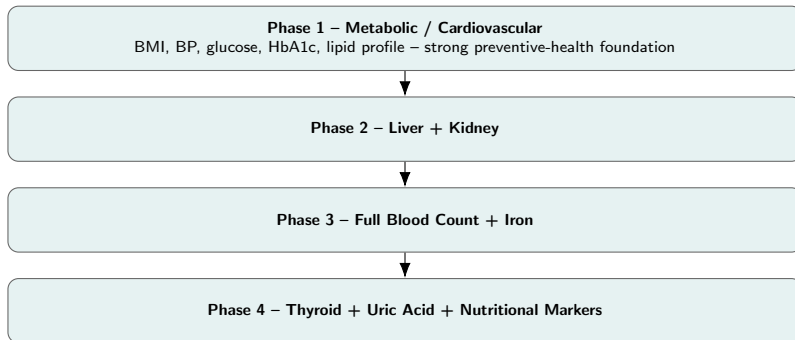
Sequence domains by preventive-health impact across four build phases.

Conclusion

Depth-first on one domain beats shallow coverage of eight.

30. Clinical Domains to Build First

Four sequential build phases



31. The Four Master Matrices Needed

Motivation

Before any code is written, the clinical logic needs to exist somewhere reviewable by non-engineers.

Result

Four spreadsheets that fully specify platform behaviour before development starts.

Method

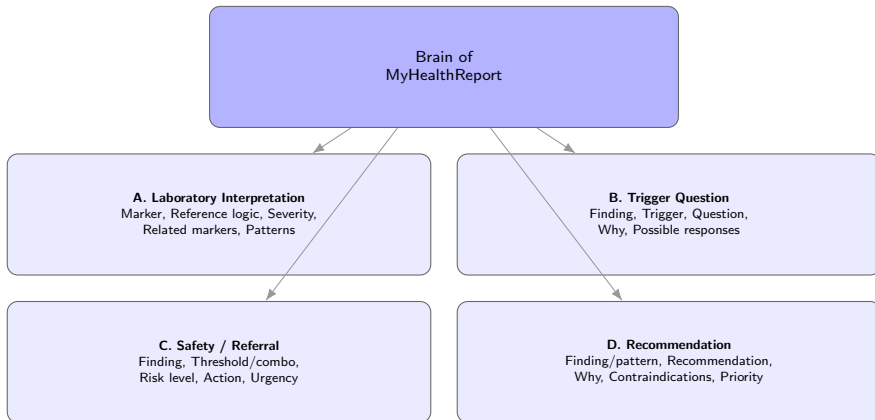
Define four matrices – Laboratory Interpretation, Trigger Question, Safety/Referral, Recommendation – each with a fixed column schema.

Conclusion

These four matrices are the brain of MyHealthReport; the software is their execution engine.

31. The Four Master Matrices Needed

Four matrices, one shared purpose



32. Version 1 Decision Flow

Motivation

Twenty modules need to be seen as one decision path, not twenty independent features.

Result

One flowchart with a single explicit branch point: *Any Safety Trigger?*

Method

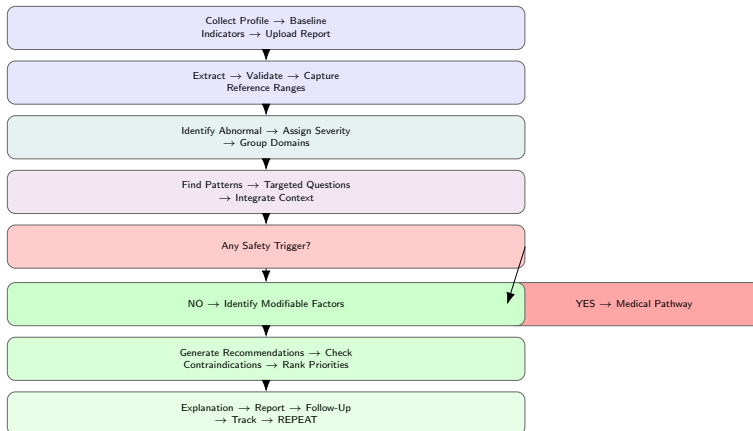
Trace a single case start-to-finish through profile, extraction, detection, questions, the safety branch, and the recommendation path.

Conclusion

Every module in this deck is a step on this one path – the safety branch is the only fork.

32. Version 1 Decision Flow

One end-to-end case, one explicit fork



33. The Identity of MyHealthReport

Motivation

Positioning the platform as “AI tells you what your blood test means” undersells and misframes it.

Result

A positioning statement – “a personalised health roadmap” – grounded in real pipeline behaviour.

Method

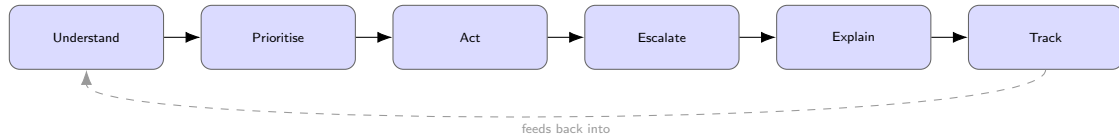
Reframe the promise around six verbs the pipeline actually performs.

Conclusion

The identity statement is not marketing copy; it is a direct re-statement of Modules 1–20.

33. The Identity of MyHealthReport

Six verbs, one cycle (Track feeds back into Understand)



34. The Central MyHealthReport Algorithm

Motivation

Thirty-four sections need to collapse into one statement a stakeholder can hold in their head.

Result

Health Interpretation, Action Plan and Safety Override as three linked equations.

Method

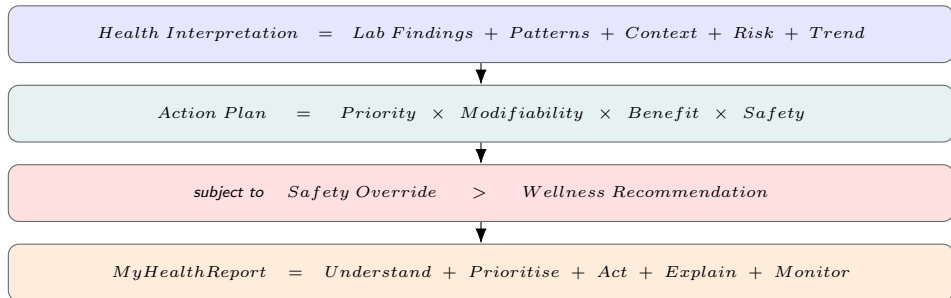
Express interpretation as a sum of five inputs, action planning as a product of four factors, subject to a safety override.

Conclusion

This is the master architecture; the Four Master Matrices (Topic 31) are the next layer beneath it.

34. The Central MyHealthReport Algorithm

Three equations, chained into one architecture



Thank You

MyHealthReport — Version 1 Framework (Expanded Edition)